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Effect of the Number of Variables on Measures of Fit in Structural Equation Modeling

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There has been relatively little systematic investigation of the effect of the number of variables on measures of model fit in structural equation modeling. There is conflicting evidence as to whether measures of fit tend to improve or decline as more variables are added to the model. We consider 3 different types of specification error: minor factors, 2-factor models, and method errors. Using a formal method based on the noncentrality parameter (NCP), we find that root mean squared error of approximation (RMSEA) tends to improve regardless of the type of specification error and that the comparative fit index (CFI) and Tucker–Lewis Index (TLI), generally, though not always, tend to worsen as the number of variables in the model increases. The formal method that we develop can be used to investigate other measures of fit and other types of misspecification.

Ever since the seminal article by Bentler and Bonett (1980), measures of fit have been an integral component of structural equation modeling. Fit indexes quantify the degree of correspondence between a hypothesized latent variable model and the data (Hu & Bentler, 1995). There has been considerable interest in both the effects of sample size and the number of parameters on measures of fit (Ding, Velicer, & Harlow, 1995; Fan, Thompson, & Yang, 1999; Hu & Bentler, 1995, 1998, 1999; Marsh, Balla, & McDonald, 1988; Marsh, Hau, Balla, & Grayson, 1998); however, there has not been a systematic investigation of the effect of the number of variables on measures of fit. To remedy this, we explicitly consider the effect of the number of variables in the model on several measures of fit. In this article, we use three approaches to understand the impact of the number of variables on selected fit indexes. First, we review the literature on the number of indicators per factor to see what effect this has on model fit. We then examine fit indexes analytically, deriving population-level fit indexes for differing numbers of indicators within particular patterns of measurement model misspecification. Third, we conduct computer simulations to evaluate our new analytic method.

First we need to clarify what we mean by “the number of variables in the model.” We are referring to the number of observed or manifest variables in the model, not the number of latent variables in the model. Consider two models, M_1 and M_2 , where M_2 contains a random subset of observed variables contained in M_1 subject to the constraint that the same latent-variable model still remains identified. Thus, M_2 contains fewer indicators per latent variable than M_1 . This article explores the effect of number of variables on measures of fit. Other issues arise when researchers vary the number of variables in the model (e.g., likelihood of convergence and improper solutions); however, these issues are not the focus of this study.

TYPES OF FIT INDEXES

Absolute fit indexes evaluate the degree to which the specified model reproduces the sample data. Some of the more commonly used absolute fit indexes include the root mean squared error of approximation (RMSEA), the Goodness-of-Fit Index, and the standardized root mean squared residual. Incremental fit indexes measure the proportionate amount of improvement in fit when a target model is compared with a more restricted, nested baseline model (Hu & Bentler, 1998). There are three types of incremental fit indexes. A Type 1 incremental fit index compares the fit function of a baseline model to the specified model. No distributional assumptions are placed on Type 1 indexes (Hu & Bentler, 1998). A drawback to Type 1 fit indexes is that they “are influenced by the badness of the null model as well as the goodness of fit of the target model” (Hu & Bentler, 1998, p. 448). Type 2 fit indexes impose additional constraints, including the assumption that the fit function of the estimated model follows a chi-square distribution with the degrees of freedom of the estimated model (Hu & Bentler, 1998). The Nonnormed Fit Index (NNFI) is the most widely used of the Type 2 incremental fit indexes. Type 3 fit indexes assume a noncentral chi-square distribution (Hu & Bentler, 1998). These noncentrality-based fit indexes include the comparative fit index (CFI) and the Relative Noncentrality Index.

In this article, we focus on three commonly used fit indexes: the CFI, a Type 3 incremental fit index; the Tucker–Lewis Index (TLI, also called the NNFI), a Type 2 incremental fit index; and the RMSEA (Browne & Cudeck, 1993; Steiger & Lind, 1980), an absolute measure of fit. The formulas for these fit indexes are as follows:

$$\begin{aligned} \text{CFI} &= 1 - [\chi^2(M) - df_M] / [\chi^2(N) - df_N] \\ \text{TLI} &= [\chi^2(N) / df_N - \chi^2(M) / df_M] / [(\chi^2(N) / df_N) - 1] \\ \text{RMSEA} &= \sqrt{[\chi^2(M) - df_M] / ((N - 1)df_M)} \end{aligned}$$

where N refers to the null model and M to the specified model (N refers to sample size). For the CFI, if the index exceeds 1 (when $\chi^2(M) < df_M$), the index is set to 1; also, for the RMSEA, if the index is negative (when $\chi^2(M) < df_M$), the index is set to zero. The approach developed in this article can be applied to other measures of fit; however, we limit our attention to these measures.

The measures of fit can be defined in terms of the noncentrality parameter (NCP) or λ . The estimate of λ of the usual maximum likelihood chi-square test is $\chi^2 - df$, where df refers to the degrees of freedom of the model. We can express the measures of fit (Rigdon, 1996) in terms of the NCP:

$$CFI = 1 - \lambda_M/\lambda_N \quad (1)$$

$$TLI = 1 - [\lambda_M/df_M]/[\lambda_N/df_N] \quad (2)$$

$$RMSEA = \sqrt{[\lambda_M/((N-1)df_M)]} \quad (3)$$

where λ_M and λ_N are the NCPs for the specified and null models, respectively. Defining the fit indexes in terms of the NCP allows us to determine the theoretical values of the fit indexes, a topic we return to later.

WHY IS THIS QUESTION IMPORTANT?

If the number of variables affected the degree of fit, then that would likely lead to practices that are undesirable. In this section, we consider what might happen if more variables led to worse fit.

If models with more variables had worse fit, practitioners would be tempted to adopt a variety of less than suboptimal strategies. For instance, some investigators might collapse across items and form item parcels, thereby reducing the number of variables in the model (Hall, Snell, & Foust, 1999). "From an empiricist-conservative philosophy of science perspective, parceling is akin to cheating because modeled data should be as close to the response of the individual as possible in order to avoid the potential imposition, or arbitrary manufacturing of a false structure" (Little, Cunningham, Shahar, & Widaman, 2002, p. 152).

Some might create a single index for each latent variable by adding or averaging all of the indicators. Alternatively, researchers might compute the index's reliability, fix the error variances to nonzero values, and estimate the model (Williams & Hazer, 1986). The problem with this approach is that it treats the error variances as known parameters rather than values that are statistically estimated. A more problematic alternative has been to treat this index as if its reliability were perfect. The failure to adjust for unreliability is a serious deficiency and leads to biased parameter estimates.

Still others might break down large models and estimated submodels that contain only a subset of the variables. For instance, if there were five constructs, each with five indicators, five different models for each of the constructs would be estimated. Thus, the models would contain 5, not 25, variables. Although this approach may be appropriate and informative for preliminary analyses, researchers are ill-advised to use the single-factor models for their final analysis, as this approach fails to detect any specification errors that occur across constructs and does not evaluate the discriminant validity of the factors.

A related problem occurs when confirmatory factor analysis is done piecemeal. For instance, in over-time data, the analysis might be done for each wave. Alternatively, when there are multiple informants (e.g., parent and child), the analysis might be done separately for each. Often it might appear that the model fits well when analyzed piecemeal, but poorly when all the parts are analyzed together. In this case, the whole is less than the sum of the parts.

If the number of variables in the model affects various measures of fit, then researchers may mistakenly trim variables out of their models to achieve acceptable fit indexes in the .90s. Marsh et al. (1998) described this temptation:

Because of the importance typically placed—rightly or wrongly—on evaluating the fit of one model against a fixed standard (e.g., the obtained χ^2 being less than its critical value, or incremental fit indices being greater than .90) this bias would naturally extinguish the possibly desirable strategy of using larger numbers of indicators. (p. 217)

They also noted that “a superficial—and erroneous—interpretation of these findings would argue that researchers should design studies with only two or three indicators per factor” (p. 216). However, models containing more items per factor were better in that they demonstrated more proper solutions, more accurate parameter estimates, and greater reliability (Marsh et al., 1998). Perhaps the poor fit in models with many variables may explain why so many published confirmatory factor analysis and structural equation modeling studies are based on only two or three variables per factor.

PREVIOUS RESEARCH

There are two different ways to examine the effect of the number of observed variables on model fit: by examining the effects of number of variables on perfectly specified models and by examining the effects of number of variables on incorrectly specified models. If the fit of an incorrectly specified model degrades as we add observed variables, this suggests that adding variables to a model increases researchers' ability to detect model misfit. This could be considered a positive result.

However, if model fit degrades as we add additional variables to a perfectly specified model, this suggests that bias exists in the measures of model fit. In this case, fit indexes would favor models with small numbers of variables, a result that has disturbing implications for the practice of structural equation modeling.

In this section, we first review previous research documenting the effects of number of variables on correctly specified models. All of these studies are simulation studies. Then we turn our attention to studies documenting the effects of the number of variables on incorrectly specified models. We review both simulations and empirical studies that provide information about the effects of the number of variables on measures of fit. We focus on the measures of fit emphasized in this article: CFI, TLI, and RMSEA.

Perfectly Specified Models

Ding et al. (1995) conducted a Monte Carlo simulation study using a perfectly specified model. They examined four levels of sample size (50, 100, 200, and 500), three levels of loading size (.5, .7, and .9), and two levels of estimation method (maximum likelihood and generalized least squares). In their simulation, four incremental fit indexes—the Bentler–Bonett or Normed Fit Index, the TLI, the Relative Noncentrality Index, and the CFI—deteriorated as the number of indicators per factor increased. For example, using maximum likelihood estimation, the average TLI decreased from 1.005 in a model with two variables per factor to .984 in a model with six variables per factor. The average CFI decreased from .995 in a model with two variables per factor to .983 in a model with six variables per factor. Ding et al. commented that it was “troubling that increasing the p/r ratio [the number of indicators per factor] did not result in an improvement in any of the fit indexes” (p. 139). This degradation in fit was weaker when the sample size was larger (see their Figure 5). In fact, with large sample sizes, all the indexes except the NNFI were very near one.

Marsh et al. (1998) conducted a simulation study of correctly specified models under a variety of conditions by varying sample size from 50 to 1,000 cases and number of indicators per factor from 2 to 12 variables in 35,000 Monte Carlo solutions. They observed a tendency in chi-square to penalize models with a large number of manifest variables. Their simulation also revealed that as the number of indicators per factor increased, the χ^2/df ratio of the specified model increased. As in the Ding et al. (1995) simulations, this increase in χ^2/df was weaker when the sample size was larger. For example, with a sample size of 50, the mean χ^2/df was .788 with 2 indicators and 1.476 with 12 indicators. However, with a sample size of 1,000, the χ^2/df was .969 with 2 indicators and 1.015 with 12 indicators. Marsh et al. concluded,

(T)his apparent decline in fit must reflect problems in standards used to evaluate fit, rather than misspecification in the approximating model. It also follows that this sys-

tematic bias in the χ^2 must also affect the behavior on a wide variety of subjective fit indexes that indices be expressed as a function of χ^2 . (p. 217)

In their Monte Carlo study, Anderson and Gerbing (1984) examined several fit indexes for models with sample sizes of 50, 100, 150, and 300. They estimated models with two to four indicators per factor, two to four factors, and loadings of .6, .9, or mixed .6 and .9 loadings. The number of indicators per factor influenced several fit indexes. As the number of indicators increased from two to four, the mean value of the TLI declined from 1.066 to .984. In general, the fit indexes indicated less fit as the number of factors in the model or the number of variables per factor increased (Anderson & Gerbing, 1991). As the sample size increased, the fit indexes improved.

These studies clearly demonstrate the effects of the number of variables on several measures of fit. There seems to be a tendency for some fit indexes to decline as the number of variables in the model increases. This trend is more pronounced at small sample sizes. Because these models were perfectly specified, there is theoretically no reason to expect measures of fit to decline as the number of variables increases. Therefore, these results suggest that there is a small amount of bias in several commonly used measures of fit.

Misspecified Models

Simulation studies. Breivik and Olsson (2001) examined the effect of the number of variables on the RMSEA and the CFI. All the models contained three indicators for each latent construct. They tested models with 3, 6, 9, and 12 latent constructs. They also added several different types of specification errors. They omitted method effects from several of the models, and they varied the number of observed variables affected by the omitted method factors. In their simulation, the RMSEA declined as the number of variables increased regardless of the source of specification error. The CFI was less affected by the number of variables. Whether the CFI tended to increase slightly or decrease slightly appeared to depend on the type of specification error.

Empirical studies. Chau and Hocevar (1995) investigated the effects of the number of measured variables on several measures of model fit. Using student-response data from the 28-item Student Evaluations of Educational Quality (Marsh, 1987) measured on 7,407 students, they estimated the same seven-factor model with two, three, and four indicators per factor. They randomly selected the indicators for the two- and three-variable runs. Because this study used an actual data set and not a simulated one, the model contained specification errors. The value of the CFI substantially decreased as the number of indicators increased from two to four when using maximum likelihood estimation (Chau & Hocevar, 1995), going from

.942 with two indicators per factor to .905 with four indicators per factor. The TLI also declined going from .906 with two indicators per factor to .890 with four indicators per factor.

Although there are inconsistencies in the results of these studies, some clear findings emerge from both the simulation and empirical studies. First, in the Breivik and Olsson study (2001), the RMSEA remained stable or declined as the number of variables increased. This suggests a trend for the RMSEA to indicate better fit as the size of the model increases. Second, the CFI and the TLI remained stable or declined as the number of variables increased. The TLI and CFI tend to exhibit worse fit as the size of the model increases. It might then be the case that the TLI and CFI react differently to the number of variables in the model than the RMSEA.

Bias in the Chi-Square Test

There is a complicating factor of bias in the chi-square test. Given that there is no specification error, the estimated chi-square is not an unbiased estimator; the amount of bias is dependent on the sample size. That is, for a fixed sample size, the average value of chi-square tends to be larger than its degrees of freedom, its expected value (see, for example, Curran, Finch, & West, 1996). The chi-square test is asymptotic and so it is exact only when the sample sizes are large. With small samples, the value of chi-square overestimates its theoretical value. What is not so well known is that this bias appears to be affected by the degrees of freedom of the test. The larger the degrees of freedom, the greater the bias appears to be (Curran, Bollen, Paxton, & Kirby, 2002). This bias in the test explains the results of those who simulated correctly specified models. These studies consistently found that as the number of variables increased, the χ^2/df ratio became larger. The effect of this bias is that fits worsens with more variables in the model.

To examine the effect of number of variables on measures of fit, we constructed a series of simulations using EQS 5.7 (Bentler, 1998). We simulated perfectly specified models and varied the number of indicators. To simplify the analyses, we estimated a one-factor model (loadings of .7 and error variances of .51) and varied the number of variables within the model. Seven different models were generated ($p = 4, 6, 10, 12, 14, 20$, and 25) at three levels of sample size ($N = 100, 200$, and $1,000$). Each condition was replicated 100 times. The results are shown in Table 1.

The number of variables has a modest effect on measures of fit. When the sample size is small, both the CFI and TLI decrease as we increase the number of variables in the model. However, the RMSEA tends to improve as we add variables to the model, especially with larger sample sizes.

For sample sizes of 100, the χ^2/df increases from .857 with 4 variables to 1.106 for 25 variables. For sample sizes of 200, the χ^2/df remains relatively constant. However, at sample sizes of 1,000, the χ^2/df actually decreases from 1.127 with 4

TABLE 1
Simulation Results: Perfectly Specified Models Single-Factor Model With p
Indicators

p	Sample Size = 100				Sample Size = 200			
	TLI	CFI	$RMSEA$	χ^2/df	TLI	CFI	$RMSEA$	χ^2/df
4	1.009	.996	.024	0.857	.998	.997	.026	1.010
6	.999	.991	.027	1.024	.998	.995	.021	1.057
10	.993	.989	.026	1.072	.997	.994	.018	1.061
12	.995	.990	.022	1.042	.997	.995	.016	1.054
14	.990	.988	.024	1.075	.998	.995	.014	1.035
20	.987	.986	.024	1.073	.995	.994	.015	1.055
25	.976	.977	.029	1.106	.995	.994	.014	1.046

p	Sample Size = 1000			
	TLI	CFI	$RMSEA$	χ^2/df
4	.999	.999	.012	1.127
6	.999	.999	.011	1.121
10	1.000	.999	.006	1.001
12	1.000	.999	.006	1.023
14	1.000	.999	.006	1.025
20	1.000	.999	.006	1.012
25	1.000	.999	.004	1.008

Note. TLI = Tucker–Lewis Index; CFI = comparative fit index; $RMSEA$ = root mean squared error of approximation.

variables to 1.008 with 25 variables. Two other trends merit attention. First, because these are perfectly specified models, the χ^2/df ratio should be approximately 1. With very small sample sizes (e.g., $N = 100$), this ratio shows greater bias upward as the number of variables increases. However, with large sample sizes, the χ^2/df approaches 1 as the number of variables increases. Second, the fit of the $RMSEA$ closely parallels the χ^2/df ratio. With $N = 100$, the $RMSEA$ worsens as more variables are added to the model. With $N = 1,000$, the $RMSEA$ improves as the number of variables increases. Therefore, models with many variables and small samples may be more inclined to experience degradation in absolute fit indexes than models with many variables and large sample sizes.

THEORETICAL VALUES OF THE MEASURES OF FIT

The review of the literature, although suggestive, is not unambiguously clear as to what the effect of number of variables has on measures of fit. Some studies show increases in measures of fit as the number of the variables in the model increases,

whereas others show the opposite, and still others show no effect. We decided to attack this problem using formal analysis by deriving theoretical formulas for measures of fit.

For each of the three measures of fit that we consider in this article, we need to derive the theoretical value of λ or NCP for the specified model and for the null model. If λ is known, we can substitute that value into the formulas for the fit indexes (see Equations 1 to 3). The usual *estimate* of λ is $\chi^2 - df$. However, the formula for the population value of λ test is

$$[N - 1][\ln|\Sigma_1| + \text{tr}(\Sigma_1^{-1}\Sigma_2) - \ln|\Sigma_2| - p] \quad (4)$$

where p is the number of variables, Σ_1 is the predicted variance–covariance matrix given the model, Σ_2 is the population covariance matrix, $\ln$ is the natural logarithm, and tr is the trace of the matrix. Note that both Σ_1 and Σ_2 are theoretical covariance matrices. (In the usual formula for the chi-square test, Σ_2 is the sample covariance matrix and Σ_1 is the predicted covariance matrix given the data.)

We seek to determine the NCP for various theoretical models. We first determine the value of λ when the specified model is the null model and when the covariance matrix is homogenous, that is, all the covariances are equal. We then consider more complicated models. We initially assume that all covariances equal the same value, which we designate as κ , the diagonal values are ones, and the specified model is the null model. Given this assumption, Σ_2 is a correlation matrix where the diagonal contains ones and the off-diagonal values equal κ . For tests of the null model, the Σ_1 matrix is an identity matrix. Because $\ln|\Sigma_1| = 0$ and because $\text{tr}(\Sigma_1^{-1}\Sigma_2) = p$, Formula 4 reduces to $-(N - 1)\ln|\Sigma_2|$ (see also Fouladi & Steiger, 1999). We now seek to determine a general formula for $|\Sigma_2|$ when all the correlations equal the same value. When $p = 2$, then $|\Sigma_2|$ equals

$$1 - \kappa^2$$

when $p = 3$, we have

$$1 - 3\kappa^2 + 2\kappa^3$$

and when $p = 4$, we have

$$1 - 6\kappa^2 + 8\kappa^3 - 3\kappa^4$$

If we factor these polynomials, the results are

$$(1 - \kappa)(1 + \kappa)$$

$$(1 - \kappa)^2(1 + 2\kappa)$$

$$(1 - \kappa)^3(1 + 3\kappa)$$

for p equal to 2, 3, and 4, respectively. The pattern is apparent, and so the general formula for a determinant when the correlations are homogenous (see also Graybill, 1983) equals

$$(1 - \kappa)^{p-1} [1 + (p-1)\kappa] \quad (5)$$

The resulting formula for λ or the NCP is

$$-(N-1)\ln[(1 - \kappa)^{p-1}(1 + (p-1)\kappa)]$$

We use this same strategy to determine the NCP for various types of specification error. In this article, the specified model is always a single-factor model. We consider three very different types of specification error: minor factors, a second factor, and method errors. To preserve symmetry, we considered models in which p is even. In the minor factors model, all the correlations are biased by a constant amount, designated as δ . In the two-factor model, the first $p/2$ variables load on one factor, the next $p/2$ variables load on a second factor, and the correlation between factors is denoted as δ . For the method-error model, each measure has a correlated error with one other measure. The diagonal of the covariance matrix contains ones and the lower-diagonal values for element c_{ij} are

$$\begin{aligned} \text{minor factors: } & \kappa + \delta \text{ (} i + j \text{ even), } \kappa - \delta \text{ (} i + j \text{ odd)} \\ \text{two factor: } & \kappa\delta \text{ (} i < p/2 + 1, j > p/2 \text{), } \kappa \text{ (all other)} \\ \text{method error: } & \kappa + \delta \text{ (} j - i = p/2 \text{), } \kappa \text{ (all other)} \end{aligned}$$

To illustrate the different types of specification error, we drew path models for each. For each model, we have one common factor, labeled as “Factor,” and six indicators, and the loadings on the major factor are all the square root of κ . All common factors have unit variance. The minor-factor model, presented in Figure 1 (note that $\beta^2 = \delta$), has a second factor called “Minor Factor.” In Figure 2, we have a two-factor model and the correlation between factors is δ . Figure 3 contains the method-error model, and the errors of each indicator are correlated δ with one other indicator.

When a single-factor model is estimated, what are the predicted elements of the variance–covariance matrix? Given maximum likelihood estimation, for all three models the diagonal values are all one and the off-diagonal values, denoted as κ_M , are as follows:

$$\begin{aligned} \text{minor factors: } & \kappa - \delta/(p-1) \\ \text{two factor: } & \kappa[(\delta + 1)p/2 - 1]/(p-1) \\ \text{method error: } & \kappa + \delta/(p-1) \end{aligned}$$

We seek to determine the NCP using Equation 4. Recall that Σ_1 is the predicted variance–covariance matrix given the model, and Σ_2 is the population covariance

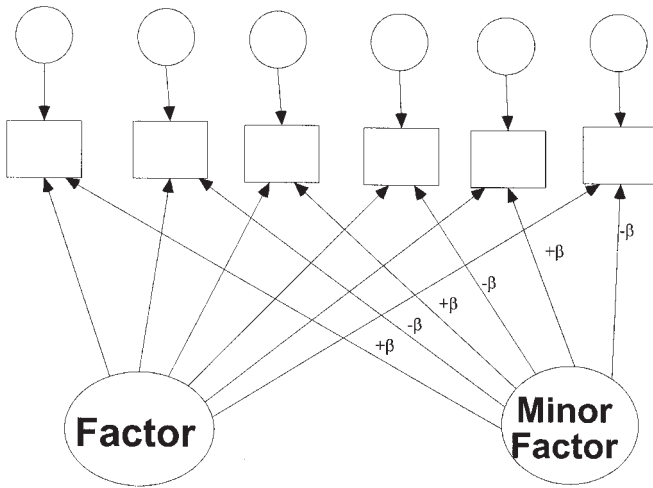


FIGURE 1 Single-factor model with a minor factor, half with positive loadings and half with negative loadings ($\beta^2 = \delta$).

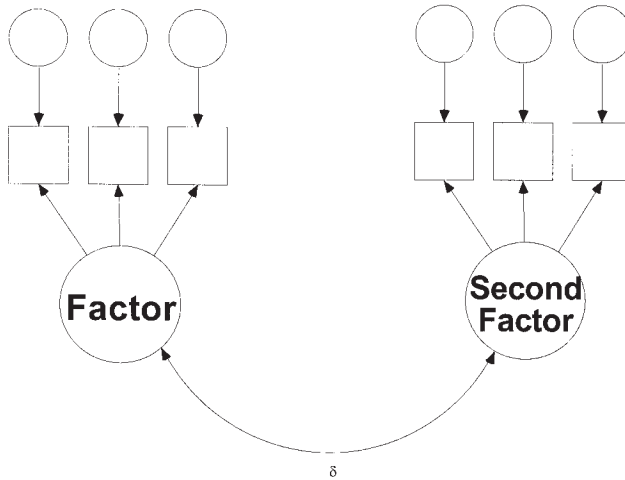
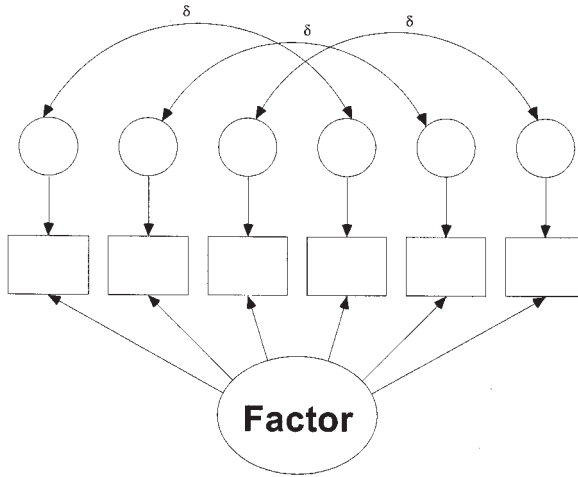


FIGURE 2 Two-factor model with correlation between factors of δ .

matrix. We can determine $\ln|\Sigma_1|$ using Formula 5 and substituting the value of κ_M for κ . The two remaining unknown terms in Formula 2 are $\text{tr}(\Sigma_1^{-1}\Sigma_2)$ and $\ln|\Sigma_2|$.

We first consider $\ln|\Sigma_2|$. The computer program Mathematica (Wolfram, 1996) was used to assist us in the derivations of the determinants of Σ_2 for various models. We determined Σ_2 for $p = 4$ and 6. We noted the pattern and then derived a general

FIGURE 3 Single-factor model with method errors whose covariance is δ .

equation for Σ_2 in terms of κ , δ , and p . As a check, we used Mathematica to reproduce the formula for $p = 8$.

The formula for the determinant of the population covariance matrix for the minor-factor model is

$$[1 - \delta + (p - 1)\kappa][1 + (p - 1)\delta - \kappa][-1 + \kappa + \delta]^{p-2} \quad (6)$$

and for the two-factor model is

$$[1 + \kappa(p/2 - 1) - \kappa\delta p/2][1 + \kappa(p/2 - 1) + \kappa\delta p/2][\kappa - 1]^{p-2} \quad (7)$$

and for the method error model is

$$[(1 + \delta - \kappa)^{p/2-1}][(1 - \delta - \kappa)^{p/2}][1 + \delta + \kappa(p - 1)] \quad (8)$$

Note that, when δ equals zero in the first and third models and when δ equals one in the second, each formula reduces to Formula 5, as it should.

We used the same approach in the derivation of the formula for $\text{tr}(\Sigma_1^{-1}\Sigma_2)$. Interestingly, we found that for all the models that we considered $\text{tr}(\Sigma_1^{-1}\Sigma_2) = p$, regardless of the form of misspecification.

We now have theoretical values for all of the ingredients of the NCP. As a check, we computed the NCP using the theoretical formulas and using a structural equation modeling computer program (EQS 5.7), and they did agree. As a further check, we conducted a series of computer simulations, and the simulation estimates of the NCP corresponded to their theoretical values. We are thus very confident of our method.

THEORETICAL VALUES OF FIT MEASURES

We have derived the values of the NCP of the null and specified models for three different types of misspecification. We can take those values and compute various measures of model fit and then vary the number of variables in the model. The measures of fit that we consider are the CFI, the TLI, and the RMSEA (formulas given previously).

We note that, for all of the models that we are considering, $\text{tr}(\Sigma_1^{-1}\Sigma_2) = p$, and so the definitional formula for λ (Formula 4) reduces to

$$(N - 1)\ln(|\Sigma_1|/|\Sigma_2|) \quad (9)$$

We denote Σ_{1N} is the predicted variance–covariance matrix for the null or independence model and Σ_{1M} for the specified model. We have previously given the formulas for the CFI, TLI, and RMSEA in terms of the NCPs (see Equations 1, 2, and 3).

We consider first the CFI. Using Formula 9, we can express λ_M/λ_N as $\ln(|\Sigma_{1M}|/|\Sigma_2|)/\ln(|\Sigma_{1N}|/|\Sigma_2|)$. Note that $|\Sigma_{1N}| = 1$ and so we have $\ln[|\Sigma_{1M}|/(|\Sigma_2|)]/\ln(1/|\Sigma_2|)$, which, because $\ln(1/|\Sigma_2|) = -\ln|\Sigma_2|$, can be shown to equal $1 - \ln[|\Sigma_{1M}|/\ln|\Sigma_2|]$. Finally, using Equation 1 and substituting the previous for λ_M/λ_N , the CFI simply equals

$$\ln|\Sigma_{1M}|/\ln|\Sigma_2| \quad (10)$$

In words, the CFI equals the ratio of the logarithm of the determinant of the predicted covariance matrix divided by the logarithm of the determinant of the covariance matrix of the data. Both of these matrices are the population matrices. By a similar logic, the TLI can be shown to equal

$$1 - [1 - (\ln|\Sigma_{1M}|/\ln|\Sigma_2|)][df(N)/df(M)] \quad (11)$$

and the RMSEA equals

$$\sqrt{[(\ln(|\Sigma_{1M}|/|\Sigma_2|))/(N - 1)df(M)]} \quad (12)$$

We can use Formula 5 to determine $|\Sigma_{1M}|$ and Formulas 6, 7, and 8 for $|\Sigma_2|$, and with these values, we can use Formulas 10, 11, and 12 to determine the values of the three fit indexes considered in this article. As a check, we ran a small computer simulation to test the adequacy of the formulas to reproduce measures of fit. We simulated a single-factor model with three types of specification error using EQS 5.7 (Bentler, 1998). We set sample size to 200 and p to 6, 12, and 30, and ran 100 trials for each of the 9 cells. As seen in Table 2, we were able to reproduce the measures of fit quite well, the largest discrepancy, ignoring sign, being only -0.0071 . The median discrepancy, again ignoring sign, is only -0.0037 for the CFI, -0.0047 for the TLI, and 0.0017 for the RMSEA. Importantly, the relation between number

TABLE 2
Fit Indexes Estimated by Simulation and by Theoretical Formulas

	<i>Number of Variables</i>		
	<i>6</i>	<i>12</i>	<i>20</i>
Minor second factor ^a			
CFI			
Simulated	.8631	.8387	.8202
Derived	.8619	.8386	.8239
Difference	.0012	.0001	-.0037
TLI			
Simulated	.7719	.8029	.8030
Derived	.7698	.8027	.8032
Difference	.0021	.0002	-.0002
RMSEA			
Simulated	.1843	.1379	.1142
Derived	.1878	.1386	.1137
Difference	-.0035	-.0007	.0007
Two-factor model ^b			
CFI			
Simulated	.9480	.9286	.9148
Derived	.9487	.9325	.9201
Difference	-.0007	-.0039	-.0053
TLI			
Simulated	.9135	.9128	.9048
Derived	.9145	.9175	.9107
Difference	-.0010	-.0047	-.0059
RMSEA			
Simulated	.0959	.0795	.0685
Derived	.1002	.0781	.0668
Difference	-.0043	.0014	.0017
Correlated errors ^c			
CFI			
Simulated	.8563	.8584	.8658
Derived	.8542	.8638	.8710
Difference	.0021	-.0054	-.0052
TLI			
Simulated	.7605	.8269	.8500
Derived	.7570	.8340	.8559
Difference	.0035	-.0071	-.0059
RMSEA			
Simulated	.2131	.1332	.0985
Derived	.2169	.1315	.0971
Difference	-.0038	.0017	.0014

Note. CFI = comparative fit index; TLI = Tucker–Lewis Index; RMSEA = root mean squared error of approximation.

^aMinor factors of loadings of +.3 and -.3. ^bFactor loading of .7 and *r* of .8 between the two factors.

^cFactor loading of .7 and a correlation of .2 between the two errors.

of variables in the model and measures of fit is the same in the simulated and derived measures of fit.

We are now ready to determine what effect the number of variables in the model has on three widely used fit indexes for three very different types of specification error.¹ For illustration purposes, we have chosen the degree of specification error to yield a CFI of .92 when $p = 8$ and $N = 200$. We have set the value of κ to .70 for all three models. The value of δ equals .0546 for the minor-factors model and .11919 for the method-error model. The value of δ for the two-factor model is .834.

As seen in Table 3, for the RMSEA, the pattern is clear regardless of the type of specification error. It declines as the number of variables in the model increases. This decline due to specification error counteracts the tendency for the χ^2/df ratio to increase in perfectly specified models (see Table 1). This result is not unique to the values in Table 3. We tested many values of κ and δ , and we always found a decline in the RMSEA as p increased. This decline in the RMSEA holds regardless of the type of misspecification and regardless of degree of misspecification. Thus, we replicate the results of Breivik and Olsson (2001). Note that the degree of decline is rather substantial. For specification error due to method factors, when p is 4, the RMSEA is .355, and when p is 40, the RMSEA is only .067. Thus, the model goes from a "very bad" to a "reasonably good" model, even though there is, in some sense, the same specification error. Although there are practical limits to p , it is interesting to note that it appears for the three patterns of specification error that we have investigated, as p approaches infinity, the RMSEA goes to zero. The decline in the RMSEA is due to the declines in the ratio of the model chi-square to its degrees of freedom. Adding variables to the model increases the degrees of freedom of the model at a faster rate than the model chi-square. (Interestingly, this decrease in the χ^2/df ratio is true of the null model.)

As seen in Table 3, the pattern for CFI depends on the source of specification error. For two types, minor factors and two factors, the CFI decreases when p increases, and that decrease is nontrivial.² However, for the method-error model, the CFI slightly increases. We think that the reason for this difference is as follows. Because the specified model predicts that all correlations take on the same value, the degree of specification error can be indexed by the degree to which the observed correlations differ from the mean correlation. For the method-error model, the proportion of correlations that have method errors declines as the number of variables increases, and so there is declining variability. However, for the other two types of specification, the variability of the correlations increases as the number of

¹The Excel program that was used for these calculations is available at <http://users.rcn.com/dakenny/TheoFit.xls>

²When we change κ and δ , the same pattern of decreasing fit generally remains for the minor-factor and two-factor models. However, when the model has extremely poor fit ($CFI < .5$), sometimes fit improved as more variables were added to the model.

TABLE 3
Theoretical Value of Fit Measures for Three Types of Specification Errors

<i>p</i>	<i>Source of Specification Errors</i>								
	<i>Minor Factors^a</i>			<i>Two Factors^b</i>			<i>Method Errors^c</i>		
	<i>TLI</i>	<i>CFI</i>	<i>RMSEA</i>	<i>TLI</i>	<i>CFI</i>	<i>RMSEA</i>	<i>TLI</i>	<i>CFI</i>	<i>RMSEA</i>
4	.826	.942	.267	.825	.942	.240	.758	.919	.355
6	.881	.928	.193	.881	.928	.174	.864	.918	.220
8	.888	.920	.168	.888	.920	.152	.888	.920	.175
10	.889	.914	.153	.889	.914	.139	.899	.921	.150
12	.889	.909	.142	.889	.909	.130	.906	.923	.133
14	.888	.905	.134	.888	.905	.122	.910	.924	.121
16	.887	.902	.127	.886	.902	.116	.913	.925	.112
20	.885	.897	.116	.884	.896	.107	.917	.926	.099
24	.883	.893	.108	.882	.892	.100	.920	.927	.089
28	.881	.890	.102	.880	.889	.094	.922	.927	.082
32	.880	.887	.096	.878	.886	.089	.923	.928	.076
36	.878	.885	.092	.877	.884	.085	.924	.928	.071
40	.877	.884	.088	.876	.882	.081	.925	.929	.067

Note. TLI = Tucker–Lewis Index; CFI = comparative fit index; RMSEA = root mean squared error of approximation.

^a $\delta = .0546$. ^b $\delta = .834$. ^c $\delta = .11919$.

variables increases. Thus, the type of specification error determines whether the CFI increases or decreases as *p* increases.

The results in Table 3 indicate that the effect of *p* on the TLI is similar to the CFI, as would be expected given that their formulas are very similar. Note how, for the minor-factor model and the two-factor model, the TLI changes direction in that it first increases and then decreases as *p* increases. The reason for this is that the formula for the TLI includes $df(N)/df(M)$, which is always greater than 1. Because the term with this multiplier is subtracted, the TLI is less than the CFI. However, note that, as *p* increases, $df(N)/df(M)$ goes to 1, and so there is curvilinearity in the TLI function. Note that the decline in the TLI is much weaker than for the CFI. For the method-error model, the TLI increases, but most of that increase occurs when *p* is small.

DISCUSSION AND CONCLUSION

We have developed a very simple formula for the theoretical value of CFI that is presented in Formula 10. That formula requires the computation of two determinants. The formula is exact for the models that we have considered, and it might well be a reasonable approximation for other cases.

Because the problem that we are considering is quite complex, it should not be surprising that there is no simple conclusion. The effect of number of variables in the model depends on a host of factors such as the type of model being estimated, the type and degree of misspecification, distributions of the variables, the sample size, the estimation method, and the specific measure of fit. We note the limitations of the formal analysis in this article. The specified model is a single-factor model, the data are assumed to be distributed multivariate normal, and maximum likelihood estimation is used.

Nonetheless, our theoretical results are in general agreement with most of the results from previous simulations and empirical studies. They show that the RMSEA seems to improve as more variables are added to the model. Depending on the type of misspecification, the fit measures for the TLI and the CFI both decline and improve as more variables are added to the model. However, very often these changes are rather small. However, in correctly specified models, the TLI and the CFI tend to demonstrate worse fit as the number of variables in the model increases, whereas the RMSEA seems to demonstrate the opposite pattern. Therefore, it appears that the CFI and the TLI do not function well with correctly specified models that include a large number of variables.

Given that the TLI and the CFI tend to decline in correctly specified models with large numbers of variables and the RMSEA tends to improve in correctly specified models with large numbers of variables, what should a researcher do? We suggest that researchers simultaneously examine the RMSEA and the CFI or TLI in models with a large number of variables. If the TLI and CFI seem slightly lower than hoped, but the RMSEA seems a bit better, then there may be no real cause for concern. However, if a model with a large number of variables has poor RMSEA and poor TLI or CFI values, that would seem to be a sign of a truly poor fitting model.

Although we considered three very different types of specification error, there are many other types of specification error. One possibility not considered in this article is the *bad apple hypothesis*. This hypothesis is that most variables in a model are *good* variables in that their covariances are nearly recovered by the model. However, some variables are *bad* variables in the sense that their covariances are not explained well by the model. If there are relatively few bad apples and so most variables are good, then having many variables in the model increases the probability that the model includes a bad apple. Thus, models with more variables would tend to fit worse than models with fewer variables.

We have used as the specified model a relatively simple specified model of a single factor. The reader might wonder what would happen if we allowed for a more complex model. In principal, the method that we have developed could be applied to a more complex model. One advantage of the single-factor model with equal loadings is that $\text{tr}(\Sigma_1^{-1}\Sigma_2)$ equals p , which greatly simplifies the formulas. However, other, more complex models have this same feature. For instance, if we were to estimate a two-factor model with the specification error that each indicator

of one factor has correlated error with one other indicator of the other factor, the equality of $\text{tr}(\Sigma_1^{-1}\Sigma_2) = p$ still holds.

All of the formulas that we have developed assume that the distributions of the variables are multivariate normal. The work by Chau and Hocevar (1995) suggested that the problem may not be so much with the fit indexes as with the estimation method of maximum likelihood. When they used robust maximum likelihood to estimate models, they found that fit measures were not affected by the number of variables in the model.

Given the widespread adoption of structural equation modeling and confirmatory factor analysis and the increasing speed and memory of computers, we are likely to see models with large numbers of variables. It would be most unfortunate to penalize researchers for estimating elaborate, theoretically interesting models with many variables. Likely this article is not the final word on this topic. However, we have developed a method that can be used to derive theoretical measures of fit.

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